1	(a)	Draw the refractive index profile and write down the necessary equation for step index and graded index fiber.	20
	(b)	Describe the working principle of fiber. The refractive index of a core is n_1 = 1.46, and cladding n_2 = 1.44. Under what conditions will light be trapped inside the core	20
	(c)	Differentiate between single mode and multimode fiber. What is the cut-off wavelength for single mode operation for a fiber that has a core with a diameter of 7 mm, a refractive index of 1.458, and a cladding of refractive index of 1.452? What is the V-number and the mode field diameter (MFD) when operating at l = 1.3 mm?	20
	(d)	Consider an SI fiber with n1=1.5 and n2=1.485 at 0.82 μm. If the core radius is 50 μm, how many modes can propagate? Repeat if the wavelength is changed to 1.2 μm. Find also the cut-off wavelength.	20
	(e)	Multimode GRIN fiber produces less modal pulse distortion than SI fiber. Why?	20

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